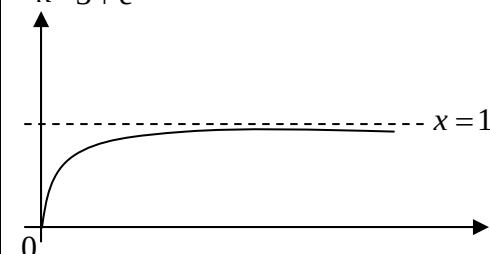


Annex B

Cover Page for College to submit with Prelim Examination Paper (NYJC – 2009 Prelim H2) :

Qn/No	Topic Set	Paper Number	Answers	Marks
1	System of Linear Equation	9740/01	86.25g, 121.25, 32.5g, \$110	4
2	Functions	9740/01	(i) $fg: x \rightarrow x+3, x \in \mathbb{R}^+$ (ii) $h: x \rightarrow \sqrt{x+7}, x > 0$ (iii) $\frac{1}{2}(1+\sqrt{29})$	6
3	Vectors	9740/01	(ii) $\sqrt{2}$	6
4	Maclaurin's	9740/01	$y = 1 + x + \frac{x^2}{2} + \frac{x^3}{3}, x + \frac{1}{6}x^3$	8
5	Recurrence relation	9740/01	(i) 1.70	9
6	Method of Difference	9740/01	(i) $\frac{1}{3} - \frac{2N}{(N+1)(N+2)}$ (ii) $\frac{2n}{(n+1)(n+2)} - \frac{1}{6}$	10
7	Graphing	9740/01	(i) -1 (ii) $y = \frac{2}{3}x$ (iii) x – intercepts are $-\frac{3k}{4} + \sqrt{\frac{9k^2}{16} - \frac{3k}{2} - 2}$ and $-\frac{3k}{4} - \sqrt{\frac{9k^2}{16} - \frac{3k}{2} - 2}$ (iv) 4	11
8	Integration (Techniques) and Volume	9740/01	(a) $\frac{1}{2}x^2 \tan^{-1}(2x^2) - \frac{1}{8} \ln(1+4x^4) + C$ (b) $\frac{\sqrt{2}}{6}$ (c) $32\pi k^5$	11
9	AP GP	9740/01	(a) $ar^{2n} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$ (b) $\frac{25}{12}, -2\ln 5$	11
10	Complex Numbers	9740/01	(i) $w = e^{i(\frac{-5k\pi}{6})}, e^{i(\frac{-\pi}{6})}, e^{i(\frac{\pi}{2})}$	12

			(ii) $\frac{1}{2} - i\frac{1}{2}\cot\left(\frac{5\pi}{12}\right), \frac{1}{2} - i\frac{1}{2}\cot\left(\frac{\pi}{12}\right), \frac{1}{2} + i\frac{1}{2}\cot\left(\frac{\pi}{4}\right)$	
11	Application of differentiation	9740/01	(a)(i) $(0,0), (0, \frac{11}{8}), (0,0), (6 \pm \sqrt{3}, 0)$ (ii) $x = -\frac{1}{8}$ (iii) $\left(66, \frac{1199}{8}\right)$ (b) $\frac{dh}{dt} = 0.143$	12
	Pure Maths			
1	Complex numbers(loci)	9740/02	$3\sin\frac{\pi}{4} = \frac{3}{\sqrt{2}}, \sqrt{8} + 2$	8
2	Differential Equations	9740/02	$x = \frac{3(1 - e^{-4t})}{3 + e^{-4t}}$ 	10
3	Area Volume	9740/02	(a) $9.87a^3$ (b) $17\frac{1}{15}\text{units}^2$	11
4	Vectors(plane)	9740/02	(i) $\mathbf{r} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -7 \\ 16 \\ 13 \end{pmatrix}, \lambda \in \mathbb{R}$ (ii) $\mathbf{r} \cdot \begin{pmatrix} 5 \\ 3 \\ -1 \end{pmatrix} = 10$ (iii) $\alpha = -8$ (iv) $\alpha = 2, \beta = 3$	13
	Statistics			
5	Regression	9740/02	The scatter diagram shows the relationship between y and x to be non-linear so it is not suitable to find the linear regression line for y on x., 0.998 As r is very close to 1, we can say that the model $\ln y = a + b \ln x$ is appropriate. $a = 3.6999, b = 0.07058$	7

6	Normal distribution	9740/02	(149.76,160.24), 0.0233, 1	9
7	testing	9740/02	(a) do not reject H_0 , 163 (b) 8.73, Distribution of iron content in substance Y is assumed to be normal	10
8	Poisson and approximation	9740/02	(i) 0.242 (ii) 0.502 (iii) 0.329 (iv) 0.858 The number of dengue fever is seasonal and thus it is not applicable over a long period of time.	11
9	Binomial/sampling	9740/02	(a) (i) 0.00707 (ii) 0.129 (b) (i) 0.975 (ii) 0.00191	11
10	P&C / Probability	9740/02	(a)265 (b)(i) 0.410, (ii) 0.0810, (iii) $\frac{9}{19}$	11